

Assignment Day - 1

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Day 1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU and both HDD and SSD components.



d. create a comparison table showing the main difference between HDD and SSD storage in terms of speed, durability, and typical use cases.

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Feature	HDD Hard disk drive	SSD Solid-state drive
Speed	slow (~50-160 MB/s). High latency due to physical disk spinning and head movement.	Fast (~500 MB/s for SATA; up to NVMe PCIe 4.0 15.0). Near times.
Durability	vulnerable to physical shocks, drops and vibrations while active due to moving mechanical parts.	Highly durable. Shock-resistant parts; survives drops easily.
Typical use cases	Bulk archival storage, Network Attached Storage (NAS), media servers and budget desktop backups.	operating system drives, gaming rigs, video editing, laptops, and fast portable drives.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM and GPU in running a game like PUBG or Free Fire on a PC.

→ * CPU (Central processing unit) :-

The CPU acts as the brain of your computer, managing game logic, physics and enemy AI. In games like PUBG or Free Fire, it calculates player movements, bullet hits, and keeps track of everything happening across the map.

* RAM (Random Access memory) :-

RAM serves as super-fast short-term memory that holds all the active game files while you play. It allows the game to instantly load textures, character skins, and nearby buildings so you do not experience sudden lag or freezing while moving.

* GPU (Graphics processing unit) :-

The GPU is dedicated entirely to rendering. GPU is dedicated entirely to visual elements like 3D landscapes, gun models, shadows and lighting effects. It takes the instructions from the CPU and turns them into smooth visual images on your monitor at high frame rates.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

→ To play the latest FIFA Sports FC smoothly at high graphic setting and 60+ FPS, here are the main hardware components,

* GPU (graphics cards / graphics processing unit) :-
Why :-

This is the main most important component for gaming. FIFA uses detailed player physics, realistic crowd lighting and high-definition pitch textures. Upgrading to a solid budget or mid-range card (like AMD RX 6600/7600) prevents frame drops during intense penalty box scrambles or cutscenes.

* Processor (CPU - central processing unit) :-
Why :-

FIFA calculates complex AI movement for 22 players on the pitch simultaneously, along with realistic ball physics. A weak CPU causes stuttering even if your graphic card is fast. A modern 6-core processor handles these calculations without lag.

* SSD (Solid State Drive) :-
Why :-

Modern FIFA games require roughly 100GB of storage space. For installing game fast NVMe SSD rather than an old mechanical Hard drive (HDD).

It's reduces patch loading times and prevents in game stutters when menu assets or cutscenes load.

* RAM (Random Access memory / System memory):
Why:-

While older games run fine on 8 GB, modern titles recommend at least 16 GB to 32 GB of RAM. Upgrading to 16 GB of DDR4 or DDR5 RAM gives your system enough breathing room to run the game smoothly for app's like Spotify.